

# Python: Standard Library Cheat Sheet

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Fundamentals of the standard library — the "batteries included" modules that ship with Python itself, no installs needed. Each section shows the everyday core of one module or task; the full APIs live at [docs.python.org](https://docs.python.org). Language topics stay on the basics and advanced sheets.

## File & Console I/O

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The basics sheet introduces `open()`; this is the fuller picture — appending, binary mode, and the standard streams.

```
import sys

# Console: print() writes to STDOUT; send errors to STDERR
print("normal output")          # -> stdout
print("something went wrong", file=sys.stderr)  # -> stderr
sys.stdout.write("raw write\n")  # print() without separators/newline logic
# input() reads a line from stdin (see the getting-started sheet)

# The standard streams ARE ordinary file objects — the same type open()
# returns, pre-opened by the interpreter. The whole file interface works,
# and they fit anywhere a file object is expected (e.g. print(file=...)).
type(sys.stdout)                # <class '_io.TextIOWrapper'> (when a terminal)
sys.stderr.write("oops\n")      # 5 -> write() returns the chars written
sys.stdout.writable()           # True   (stdin: read-only; stdout/err: write-only)
sys.stdin.readable()            # True
# text = sys.stdin.read()       # read stdin until EOF — would sit waiting here
# for line in sys.stdin:        # ...or iterate it line by line, like any file
# Never close() them — the interpreter owns their lifetime.

# Text files: pass an explicit encoding, and let `with` close the file
with open("notes.txt", "w", encoding="utf-8") as f:  # w = write (TRUNCATES)
    f.write("first line\n")

with open("notes.txt", "a", encoding="utf-8") as f:  # a = append to the end
    f.write("second line\n")

with open("notes.txt", encoding="utf-8") as f:      # mode "r" is the default
    content = f.read()
content      # 'first line\nsecond line\n'

# Read line by line (memory-friendly: never loads the whole file)
with open("notes.txt", encoding="utf-8") as f:
    for line in f:
        print(line.rstrip())    # first line / second line

# Binary files: add "b" to the mode -> bytes in, bytes out, no encoding
# (the bytes type itself is covered on the basics sheet)
with open("blob.bin", "wb") as f:
    f.write(b"\x00\x01\xff")

with open("blob.bin", "rb") as f:
    data = f.read()
data      # b'\x00\x01\xff'
len(data) # 3

# Modes cheat: r read | w write (truncate!) | a append | x create, fail
```

```
# if it exists | +b modifiers combine: "rb", "a+", "xb"...

# Newlines: TEXT mode translates them, BINARY mode never touches them.
# Reading text converts \r\n (Windows) and \r to \n on ANY OS; writing
# text converts \n to the platform's ending (\r\n on Windows only).
with open("crlf.txt", "wb") as f:
    f.write(b"one\r\ntwo\n")          # raw bytes with a Windows line

with open("crlf.txt", encoding="utf-8") as f:
    f.read()                          # 'one\ntwo\n'      -> \r\n silently became \n

with open("crlf.txt", "rb") as f:
    f.read()                          # b'one\r\ntwo\n'   -> binary: untouched

# newline="" switches the translation OFF in text mode (csv needs this)
with open("crlf.txt", encoding="utf-8", newline="") as f:
    f.read()                          # 'one\r\ntwo\n'

# The platform's own ending lives in os.linesep: '\n' here, '\r\n' on
# Windows – but code rarely needs it: just write '\n' and let text
# mode translate.
```

## Command-Line Arguments

```
import sys

# sys.argv holds the arguments as a list of STRINGS; [0] is the script.
# Running:  python myscript.py input.txt --fast
# gives:    sys.argv == ['myscript.py', 'input.txt', '--fast']

# argparse: declare the arguments, get parsing, validation, and --help
import argparse

parser = argparse.ArgumentParser(description="Process a file")
parser.add_argument("path")                # required positional
parser.add_argument("--fast", action="store_true") # boolean flag
parser.add_argument("--level", type=int, default=1) # typed, with default

# In a real script: args = parser.parse_args() (reads sys.argv)
# An explicit list also works – used here to keep the example runnable:
args = parser.parse_args(["input.txt", "--fast", "--level", "3"])
args.path          # 'input.txt'
args.fast          # True
args.level         # 3 -> already an int, thanks to type=int

# Bad input never reaches your code: argparse prints usage and exits
# parser.parse_args([])          # error: the following arguments are
#                               # required: path -> SystemExit(2)
# parser.parse_args(["-h"])     # prints full help and exits
```

## Environment Variables

```
import os

# os.environ is a dict-like view of the process environment
# os.environ["HOME"]          # e.g. '/Users/alice' – varies per machine
os.environ.get("API_KEY")     # None -> missing key, no error (like dict.get)
os.environ.get("API_KEY", "") # "" -> or supply your own default
```

```

# os.environ["API_KEY"]          # KeyError if not set ([] access raises)

# Setting: values must be STRINGS; child processes inherit them
os.environ["APP_MODE"] = "debug"
os.environ["APP_MODE"]          # 'debug'
# os.environ["PORT"] = 8000      # TypeError: str expected, not int
os.environ["PORT"] = "8000"     # numbers go in (and come out) as strings
int(os.environ["PORT"])         # 8000 -> convert on the way out

del os.environ["APP_MODE"]      # unset
"APP_MODE" in os.environ        # False

# Changes affect THIS process and its children only – never the shell
# that launched it; everything reverts when the process exits.

```

## JSON

```

import json

# Python -> JSON string: dumps ("dump to string"). The typical input is
# a DICT (any mix of the mapped types below works):
data = {"name": "Alice", "age": 30, "tags": ["admin"], "active": True}
json.dumps(data)    # '{"name": "Alice", "age": 30, "tags": ["admin"], "active": true}'

# JSON string -> Python: loads. A JSON object comes back as a real DICT
# (a top-level JSON array would come back as a list):
json.loads('{"x": 1, "ok": true, "note": null}')
# {'x': 1, 'ok': True, 'note': None}    -> a dict; true/null became True/None

# Any mapped type works at the TOP level too – e.g. a list
json.dumps([1, "two", True])    # '[1, "two", true]'
json.loads("[1, 2, 3]")        # [1, 2, 3]

# Mapping: dict<->object, list/tuple<->array, str<->string, int/float
# <->number, True/False<->true/false, None<->null. Sets don't fit:
# json.dumps({1, 2})    # TypeError: Object of type set is not JSON serializable

# Class instances don't serialize by themselves either...
class User:
    def __init__(self, name, age):
        self.name = name
        self.age = age

u = User("Alice", 30)
# json.dumps(u)          # TypeError: Object of type User is not
#                        # JSON serializable
# ...but their attribute dict does – vars(u) is u.__dict__ (see the
# advanced sheet's Introspection section):
json.dumps(vars(u))        # '{"name": "Alice", "age": 30}'
json.dumps(u, default=vars) # same result – and default= also covers
                           # objects NESTED anywhere in the data

# Pretty-print with indent
print(json.dumps({"a": 1, "b": [2, 3]}, indent=2))
# {
#   "a": 1,
#   "b": [
#     2,
#     3
#   ]
# }

```

```
# }

# Files: dump/load (no s) work on file objects directly
with open("config.json", "w", encoding="utf-8") as f:
    json.dump(data, f, indent=2)

with open("config.json", encoding="utf-8") as f:
    loaded = json.load(f)
loaded == data      # True -> clean round-trip

# Non-ASCII is escaped by default; ensure_ascii=False keeps it readable
json.dumps({"city": "città"})          # '{"city": "citt\\u00e0"}'
json.dumps({"city": "città"}, ensure_ascii=False) # '{"city": "città"}'

# GOTCHA: JSON object keys are ALWAYS strings - non-string keys convert
json.loads(json.dumps({1: "one"}))    # {'1': 'one'} -> the int key became '1'
```

## Base64

Base64 represents arbitrary **bytes** as plain ASCII text — the standard way to move binary data through text-only channels (JSON, URLs, email). It is an encoding, **not encryption**: anyone can decode it, so it protects nothing.

```
import base64

# bytes in -> bytes out (the bytes type is on the basics sheet)
raw = b"hi there"
encoded = base64.b64encode(raw)      # b'aGkgdGhlcmU='
base64.b64decode(encoded)            # b'hi there' -> clean round-trip

# Strings need the str<->bytes hop on BOTH ends
b64 = base64.b64encode("città".encode("utf-8")).decode("ascii")
b64                                     # 'Y2l0dMOg' -> a plain str now
base64.b64decode(b64).decode("utf-8") # 'città'

# URL-safe variant: substitutes _ for +/ (safe in URLs and filenames)
base64.b64encode(bytes([251, 255]))   # b'+/8='
base64.urlsafe_b64encode(bytes([251, 255])) # b'/_8='
```

## Hashing

A hash is a fixed-length fingerprint of some bytes: one-way (no decoding back), deterministic, and any tiny change produces a completely different digest. Uses: integrity checks, deduplication, cache keys.

```
import hashlib

# The raw result is BYTES (32 for sha256): digest().hexdigest() is the
# SAME value as hex text - 2 chars per byte, 64 total - which is what
# you print, store, and compare. hexdigest() == digest().hex()
hashlib.sha256(b"hi there").digest()[0:4] # b'\x9b\x96\xa1\xfe' (raw)
h = hashlib.sha256(b"hi there").hexdigest()
h                                     # '9b96a1fe1d548cbbc960cc6a0286668fd74a763667b06366fb2324269fcabaa4'
len(h)                               # 64 -> always, regardless of the input's size
# (Below, [0:16] only shortens output for THIS sheet - no other meaning.)

# The avalanche effect: one changed byte, entirely different digest
hashlib.sha256(b"hi therE").hexdigest()[0:16] # '6b2bf6242cbce8a0'
```

```

# Other algorithms, same interface. md5/sha1 are BROKEN for security -
# still fine for checksums and dedup, never for anything an attacker
# could exploit. Prefer sha256.
hashlib.md5(b"hi there").hexdigest()      # 'fd33e2e8ad3cb1bdd3ea8f5633fcf5c7'

# Feed data in chunks with update() - e.g. hashing a big file
inc = hashlib.sha256()
inc.update(b"hi ")
inc.update(b"there")
inc.hexdigest() == h      # True -> same as hashing it in one go

# PASSWORDS are a special case: plain sha256 is too fast to be safe.
# Use a slow, salted derivation like pbkdf2 (or scrypt):
dk = hashlib.pbkdf2_hmac("sha256", b"password", b"salt", 100_000)
dk.hex()[:16]             # '0394a2ede332c9a1'

# secrets: cryptographically strong random tokens (never use random!)
import secrets
secrets.token_hex(8)       # e.g. '162bae35215e0d6e' - new every call
secrets.token_urlsafe(8)   # e.g. '02k0llgqkIs'

```

## Dates & Times (datetime)

```

from datetime import date, datetime, timedelta

# The current moment - values obviously change every run:
today = date.today()      # a date:      e.g. date(2026, 7, 12)
now = datetime.now()      # a datetime:  date + time of day

# Fixed values are built by component
d = date(2026, 7, 12)
dt = datetime(2026, 7, 12, 9, 30)      # year, month, day, hour, minute
d.year, d.month, d.day                # (2026, 7, 12)
d.weekday()                          # 6 -> Monday is 0 ... Sunday is 6

# Arithmetic: timedelta is a DURATION; +/- shift dates, - gives gaps
d + timedelta(days=1)                # date(2026, 7, 13)
d + timedelta(weeks=2)               # date(2026, 7, 26)
deadline = datetime(2026, 12, 31, 23, 59)
(deadline - dt).days                # 172 -> subtraction yields a timedelta
d < date(2026, 12, 25)               # True -> comparisons just work

# Object -> string: strftime ("string format time")
dt.strftime("%Y-%m-%d")               # '2026-07-12'
dt.strftime("%d/%m/%Y %H:%M")        # '12/07/2026 09:30'
dt.isoformat()                       # '2026-07-12T09:30:00'

# String -> object: strptime ("string parse time") - the reverse
datetime.strptime("12/07/2026 09:30", "%d/%m/%Y %H:%M")
# datetime(2026, 7, 12, 9, 30)
datetime.fromisoformat("2026-07-12T09:30:00") # shortcut for ISO strings

# All of the above are NAIVE (no timezone attached). For aware objects:
from datetime import timezone
utc_now = datetime.now(timezone.utc)   # tzinfo=UTC -> comparable across
                                         # zones; prefer aware in real apps

```

## Regular Expressions (`re`)

```
import re

text = "Order 66 shipped on 2026-07-12 to Alice"

# Write patterns as RAW strings (r"...") so \d survives untouched
re.search(r"\d+", text)      # <re.Match object; span=(6, 8), match='66'>
re.search(r"\d+", text).group() # '66' -> first match, anywhere
re.match(r"\d+", text)       # None -> match() anchors at the START only
re.search(r"xyz", text)      # None -> no match is None: always check!

re.findall(r"\d+", text)     # ['66', '2026', '07', '12'] -> all matches
re.sub(r"\d", "#", text)     # 'Order ## shipped on ####-##-## to Alice'

# Groups: parentheses capture parts of the match
m = re.search(r"(\d{4})-(\d{2})-(\d{2})", text)
m.group(0)                  # '2026-07-12' -> the whole match
m.group(1)                  # '2026' -> first captured group
m.groups()                  # ('2026', '07', '12')

# Named groups read better
m = re.search(r"(?P<year>\d{4})-(?P<month>\d{2})", text)
m.group("year")             # '2026'

# compile() the pattern once when reusing it; flags tweak the rules
word = re.compile(r"alice", re.IGNORECASE)
word.search(text).group()   # 'Alice'

# Mini pattern reference:
# \d digit    \w word char    \s whitespace    . any char
# + one or more    * zero or more    ? optional    {n} exactly n
# ^ start        $ end    [abc] char set    (a|b) alternation
```

## Random Numbers (`random`)

Pseudo-random numbers for simulations, games, sampling. **Not for security** — tokens and passwords need secrets (see Hashing above).

```
import random

# In normal use, do NOT call seed(): Python seeds itself from OS
# entropy on first use, so every run gets a fresh, unpredictable
# sequence. Fix the seed ONLY when you WANT repeatability — tests,
# experiments, demos: same seed -> same numbers, forever (it is how
# the outputs below can be exact). A fixed seed is predictable by
# design; seed() with no argument re-seeds unpredictably from the OS.
random.seed(42)

random.random()             # 0.6394267984578837 -> float in [0.0, 1.0)
random.randint(1, 6)        # 1 -> int, BOTH ends included
random.choice(["rock", "paper", "scissors"]) # 'scissors' -> one element

# sample(population, k): k DISTINCT picks, no repeats — a lottery draw.
# The population can be any sequence; range(1, 50) is the ints 1..49
random.sample(range(1, 50), 3) # [18, 16, 15]
```

```

# uniform(a, b): one float anywhere between a and b (ends included) -
# the pick-from-a-range version of random(), which is locked to [0, 1)
random.uniform(1, 10)      # 2.255841356726295

# shuffle() reorders a list IN PLACE (mutates it) and returns None,
# so never write cards = random.shuffle(cards) - cards would be None!
cards = [1, 2, 3, 4, 5]
random.shuffle(cards)
cards                      # [4, 2, 3, 5, 1] -> same list, new order

# Want a shuffled COPY instead? Sample the whole list:
random.sample(cards, len(cards)) # [4, 1, 5, 3, 2] -> new list
cards                          # [4, 2, 3, 5, 1] -> original untouched

```

## Specialized Containers (collections)

Four upgrades over the built-in collections, each solving one recurring chore.

```

from collections import Counter, defaultdict, namedtuple, deque

# Counter: tallies anything iterable - the "counting dict"
votes = Counter(["yes", "no", "yes", "yes", "abstain"])
votes                      # Counter({'yes': 3, 'no': 1, 'abstain': 1})
votes["yes"]               # 3
votes["missing"]          # 0 -> missing counts are 0, never a KeyError
votes.most_common(2)       # [('yes', 3), ('no', 1)] -> top-k, ready sorted
Counter("mississippi").most_common(2) # [('i', 4), ('s', 4)]

# Counters are mutable: increment freely (missing keys start at 0),
# feed in more items with update(), even add two Counters together
votes["late"] += 1        # works although "late" wasn't there
votes.update(["no", "no"])
votes                     # Counter({'yes': 3, 'no': 3, 'abstain': 1, 'late': 1})
Counter("aab") + Counter("abb") # Counter({'a': 3, 'b': 3})

# defaultdict: a dict that CREATES the value on first access, using the
# factory you give it - kills the "if key not in d: d[key] = []" dance
groups = defaultdict(list) # factory list -> default is a new []
groups["fruit"].append("apple") # no KeyError: [] appeared, then append
groups["fruit"].append("pear")
dict(groups)                # {'fruit': ['apple', 'pear']}

# namedtuple: a tuple with NAMED fields - a lightweight, immutable record
Point = namedtuple("Point", ["x", "y"])
p = Point(3, 4)
p                           # Point(x=3, y=4) -> readable repr for free
p.x                        # 3 -> access by name...
p[0]                       # 3 -> ...but it's still a tuple (unpacking works)
# (need mutability, defaults, methods? -> dataclasses, advanced sheet)

# deque ("deck"): double-ended queue - fast appends/pops at BOTH ends,
# where lists are slow on the left
dq = deque([2, 3])
dq.appendleft(1)           # deque([1, 2, 3])
dq.append(4)               # deque([1, 2, 3, 4])
dq.popleft()               # 1 -> O(1); list.pop(0) shifts everything
dq                         # deque([2, 3, 4])
deque([1, 2, 3, 4], maxlen=3) # deque([2, 3, 4], maxlen=3)
#                           -> bounded: keeps only the LAST 3 (rolling window)

```

```
# Stack and queue? No dedicated container needed:
stack = [1, 2]           # a STACK (LIFO) is just a list...
stack.append(3)          # push
stack.pop()              # 3 -> pop; both O(1) at the right end
q = deque([1, 2])        # a QUEUE (FIFO) is a deque...
q.append(3)               # enqueue at the right
q.popleft()              # 1 -> dequeue at the left (never list.pop(0)!)
# queue.Queue/LifoQueue exist too, but they're THREAD-SAFE pipes for
# producer/consumer concurrency, not everyday containers. Priority
# queues: heapq.
```

## Running External Commands (`subprocess`)

```
import subprocess
import sys

# run() starts a program and waits for it. Pass the command as a LIST:
# no shell parsing, no quoting headaches, no injection risk.
# sys.executable is a string: the absolute path of the interpreter
# currently running (e.g. '/usr/local/bin/python3.14'). Launching it
# as the child means "this exact same Python" - version and venv
# included - so these examples run anywhere. The pattern is the same
# for any program: ["git", "status"], ["ls", "-la"], ...
result = subprocess.run(
    [sys.executable, "-c", "print('hi from a child process')"],
    capture_output=True,      # collect stdout/stderr instead of inheriting
    text=True,                # decode bytes -> str
)
result.returncode            # 0 -> the process's exit code; 0 means success
result.stdout                # 'hi from a child process\n'
result.stderr                # ''

# A failing command does NOT raise by default - inspect returncode...
bad = subprocess.run([sys.executable, "-c", "raise SystemExit(3)"])
bad.returncode               # 3
# ...or pass check=True to turn failures into exceptions:
# subprocess.run(..., check=True)
# -> CalledProcessError: Command '['...']' returned non-zero exit status 3.

# The command must be a REAL executable on PATH, and availability
# varies by OS: dir is a binary on Linux, missing on macOS (use ls),
# and a cmd.exe BUILT-IN on Windows - built-ins need their shell:
# subprocess.run(["dir"])      # macOS: FileNotFoundError
# subprocess.run(["cmd", "/c", "dir"]) # Windows: the way to a built-in

# Avoid shell=True (one big string parsed by the shell): it reintroduces
# exactly the quoting and injection problems the list form avoids.
```